[GB10] Xavier Glorot and Yoshua Bengio. Understanding the difficulty of training deep feedforward neural networks. In Yee Whye Teh and Mike Titterington, editors, *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, volume 9 of *Proceedings of Machine Learning Research*, pages 249–256, Chia Laguna Resort, Sardinia, Italy, 13–15 May 2010. PMLR.

[GD23] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.

[GJB+25] Noah Golowich, Samy Jelassi, David Brandfonbrener, Sham M Kakade, and Eran Malach. The role of sparsity for length generalization in transformers. *arXiv preprint arXiv:2502.16792*, 2025.

[GJM+20] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.

[GTLV22] Shivam Garg, Dimitris Tsipras, Percy S Liang, and Gregory Valiant. What can transformers learn in-context? a case study of simple function classes. *Advances in neural information processing systems*, 35:30583–30598, 2022.

[JdDE+23] Samy Jelassi, Stéphane d'Ascoli, Carles Domingo-Enrich, Yuhuai Wu, Yuanzhi Li, and François Charton. Length generalization in arithmetic transformers. *arXiv preprint arXiv:2306.15400*, 2023.

[KB14] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

[LAG+22] Bingbin Liu, Jordan T Ash, Surbhi Goel, Akshay Krishnamurthy, and Cyril Zhang. Transformers learn shortcuts to automata. *arXiv preprint arXiv:2210.10749*, 2022.

[LH19] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

[LLR23] Yuchen Li, Yuanzhi Li, and Andrej Risteski. How do transformers learn topic structure: Towards a mechanistic understanding. In *International Conference on Machine Learning*, 2023.

[LLZV+25] Yue M Lu, Mary Letey, Jacob A Zavatone-Veth, Anindita Maiti, and Cengiz Pehlevan. Asymptotic theory of in-context learning by linear attention. *Proceedings of the National Academy of Sciences*, 122(28):e2502599122, 2025.

[LZHO25] Mingchen Li, Xuechen Zhang, Yixiao Huang, and Samet Oymak. On the power of convolution-augmented transformer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[Man79] O.L. Mangasarian. Uniqueness of solution in linear programming. *Linear Algebra and its Applications*, 25:151–162, 1979.

[MLH+22] Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. Rethinking the role of demonstrations: What makes in-context learning work? *arXiv preprint arXiv:2202.12837*, 2022.

[MPL19] R Thomas McCoy, Ellie Pavlick, and Tal Linzen. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. *arXiv preprint arXiv:1902.01007*, 2019.

[NDL24] Eshaan Nichani, Alex Damian, and Jason D Lee. How transformers learn causal structure with gradient descent. In *International Conference on Machine Learning*, 2024.

[NSO+25] Naoki Nishikawa, Yujin Song, Kazusato Oko, Denny Wu, and Taiji Suzuki. Nonlinear transformers can perform inference-time feature learning. In *Forty-second International Conference on Machine Learning*, 2025.

[OEN+22] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context learning and induction heads. *Transformer Circuits Thread*, 2022.

[ORST23] Samet Oymak, Ankit Singh Rawat, Mahdi Soltanolkotabi, and Christos Thrampoulidis. On the role of attention in prompt-tuning. In *International Conference on Machine Learning*, 2023.

[OSSW24] Kazusato Oko, Yujin Song, Taiji Suzuki, and Denny Wu. Pretrained transformer efficiently learns low-dimensional target functions in-context. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 77316–77365. Curran Associates, Inc., 2024.

[PAA+23] Bo Peng, Eric Alcaide, Quentin Anthony, Alon Albalak, Samuel Arcadinho, Stella Biderman, Huanqi Cao, Xin Cheng, Michael Chung, Matteo Grella, et al. Rwkv: Reinventing rnns for the transformer era. *arXiv preprint arXiv:2305.13048*, 2023.

[Red24] Gautam Reddy. The mechanistic basis of data dependence and abrupt learning in an in-context classification task. In *International Conference on Learning Representations (ICLR)*, 2024.

[RLIGS22] Yasaman Razeghi, Robert L Logan IV, Matt Gardner, and Sameer Singh. Impact of pretraining term frequencies on few-shot reasoning. *arXiv preprint arXiv:2202.07206*, 2022.

[RPCG23] Allan Raventós, Mansheej Paul, Feng Chen, and Surya Ganguli. Pretraining task diversity and the emergence of non-bayesian in-context learning for regression. *Advances in neural information processing systems*, 36:14228–14246, 2023.

[SHT24] Clayton Sanford, Daniel Hsu, and Matus Telgarsky. One-layer transformers fail to solve the induction heads task. *arXiv preprint arXiv:2408.14332*, 2024.

[VSP+17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.

[VTM+19] Elena Voita, David Talbot, Fedor Moiseev, Rico Sennrich, and Ivan Titov. Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned. *arXiv preprint arXiv:1905.09418*, 2019.

[WNB+25] Zixuan Wang, Eshaan Nichani, Alberto Bietti, Alex Damian, Daniel Hsu, Jason D Lee, and Denny Wu. Learning compositional functions with transformers from easy-to-hard data. *arXiv preprint arXiv:2505.23683*, 2025.

[WS24] Shuo Wang and Issei Sato. Understanding knowledge hijack mechanism in in-context learning through associative memory, 2024.

[ZBB+22] Yi Zhang, Arturs Backurs, Sébastien Bubeck, Ronen Eldan, Suriya Gunasekar, and Tal Wagner. Unveiling transformers with lego: a synthetic reasoning task. *arXiv preprint arXiv:2206.04301*, 2022.

[ZBL+23] Hattie Zhou, Arwen Bradley, Etai Littwin, Noam Razin, Omid Saremi, Josh Susskind, Samy Bengio, and Preetum Nakkiran. What algorithms can transformers learn? a study in length generalization. *arXiv preprint arXiv:2310.16028*, 2023.

A Preliminaries

Throughout the paper, $O(\cdot)$ notation is taken with respect to the vocabulary size N and we assume the following scaling:

Assumption 1 (Scaling between problem parameters)**.** *Number of triggers N_{trg} satisfies $N_{\mathrm{trg}} = o(N^{1/3})$. Context length T satisfies $T = o(N/N_{\mathrm{trg}}^2)$ almost surely.*

We also assume the following:

Assumption 2. *We assume $\ell \geq 4$ almost surely at pretraining and $L = \mathrm{poly}(N)$.*

$\mathbb{I}(A)$ denotes the indicator function of the event A, that is, $\mathbb{I}(A) = 1$ if A holds, and 0 otherwise. For $a \leq b$, $\mathbf{1}_{a:b}$ denotes the vector whose $a, a+1, \ldots, b$-th entries are one and others are zero. For a matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, we use the notation $\boldsymbol{A}[:, I:J]$ to denote the submatrix consisting of all rows and columns from index I to J. For an arbitrary matrix $\boldsymbol{A}$, $\lambda_i(\boldsymbol{A})$ is the i-th largest eigenvalue of $\boldsymbol{A}$.

B Infinite Sample Analysis

In this section, we analyze Algorithm 1 with infinite sample size. Recall that we defined

$$f_{\mathrm{TF}}(\boldsymbol{X}_{1:t}; \boldsymbol{W}_{KQ}, \boldsymbol{W}_V) = \boldsymbol{W}_V \boldsymbol{X}_{1:t} \operatorname{Softmax}(\boldsymbol{X}_{1:t}^\top \boldsymbol{W}_{KQ} \boldsymbol{x}_t) \in \mathbb{R}^N$$

and next token prediction loss

$$\mathcal{L}(\boldsymbol{X}_{1:T^{(i)}}^{(i)}; \boldsymbol{W}_{KQ}, \boldsymbol{W}_V) = \mathrm{CrossEntropy}(\boldsymbol{e}_{z_{T^{(i)}+1}}, \operatorname{Softmax}(f_{\mathrm{TF}}(\boldsymbol{X}_{1:T^{(i)}}; \boldsymbol{W}_{KQ}, \boldsymbol{W}_V))).$$

For simplicity, we denote the population loss as $\bar{\mathcal{L}}(\boldsymbol{W}_{KQ}, \boldsymbol{W}_V) := \mathbb{E}_T[\mathbb{E}_{\boldsymbol{X}_{1:T}}[\mathcal{L}(\boldsymbol{X}_{1:T}; \boldsymbol{W}_{KQ}, \boldsymbol{W}_V)]]$.

B.1 Population Gradient of $\mathbf{W}_V$

Note that $\operatorname{Softmax}(\boldsymbol{X}_{1:T}^\top \boldsymbol{W}_{KQ} \boldsymbol{x}_T) = [1/T, \ldots, 1/T]^\top$ and $\operatorname{Softmax}(f_{\mathrm{TF}}(\boldsymbol{X}_{1:t}; \boldsymbol{W}_{KQ}, \boldsymbol{W}_V)) = [1/N, \ldots, 1/N]^\top$ is satisfied at initialization, for any $\boldsymbol{X}_{1:t}$. From [BCB+23, Lemma 1], the population loss can be calculated as

$$\begin{aligned}
&\nabla_{\boldsymbol{W}_V} \bar{\mathcal{L}}(\boldsymbol{W}_{KQ}, \boldsymbol{W}_V) \\
=&\frac{1}{N} \sum_{k=1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\boldsymbol{x}_t]^\top \right] - \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\mathbb{I}(z_{T+1} = k) \boldsymbol{x}_t^\top] \right]. \\
=&\frac{1}{N} \sum_{k=1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\boldsymbol{x}_t]^\top \right] - \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{e}_k \mathbb{E}_T \left[\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\boldsymbol{x}_t | z_{T+1} = k]^\top \right].
\end{aligned}$$

We conduct block-wise calculation for the population gradient: let

$$\boldsymbol{W}_V = \left[\boldsymbol{W}_V^{(1)}, \boldsymbol{W}_V^{(2)}, \boldsymbol{W}_V^{(3)} \right]$$

and let

$$\boldsymbol{W}_V^* = \left[\boldsymbol{W}_V^{*,(1)}, \boldsymbol{W}_V^{*,(2)}, \boldsymbol{W}_V^{*,(3)} \right]$$

be $\boldsymbol{W}_V$ after one GD step, i.e., $\boldsymbol{W}_V^* = -\eta_V \nabla_{\boldsymbol{W}_V} \bar{\mathcal{L}}(\boldsymbol{W}_{KQ}, \boldsymbol{W}_V)$, where $\boldsymbol{W}_V^{(1)} \in \mathbb{R}^{N \times D}$, $\boldsymbol{W}_V^{(2)}, \boldsymbol{W}_V^{(3)} \in \mathbb{R}^{N \times N}$. In this section we show the following, using the rescaling $\eta_V = N \tilde{\eta_V}$ for notation simplicity.